Transformers

$$o = \text{Attention}(q, k, v)$$

(query, keys, values)

$$\alpha_i = \frac{\exp\left(\frac{q^T k_i}{\sqrt{d}}\right)}{\sum_{j=1}^{T} \exp\left(\frac{q^T k_j}{\sqrt{d}}\right)}, \quad i = 1, \ldots, T$$

$$o = \sum_{i=1}^{T} \alpha_i v_i$$

Seq2Seq

$q = \tilde{h}_t$

$k = h_1, \ldots, h_T$

$v = h_1, \ldots, h_T$

$$\underset{\widetilde{T}\times H}{O} = Att\left(\underset{\widetilde{T}\times H}{Q}, \underset{\widetilde{T}\times H}{K}, \underset{T\times H}{V}\right)$$

$$O = \underset{\text{row-wise}}{Softmax}\left(\frac{\overset{\widetilde{T}\times T}{Q K^T}}{\sqrt{H}}\right) V$$

$$Q \in \mathbb{R}^{\widetilde{T}\times H_{head}} \qquad K \in \mathbb{R}^{T\times H_{head}} \qquad QK^T \to \widetilde{T}\cdot T \cdot H_{head}$$

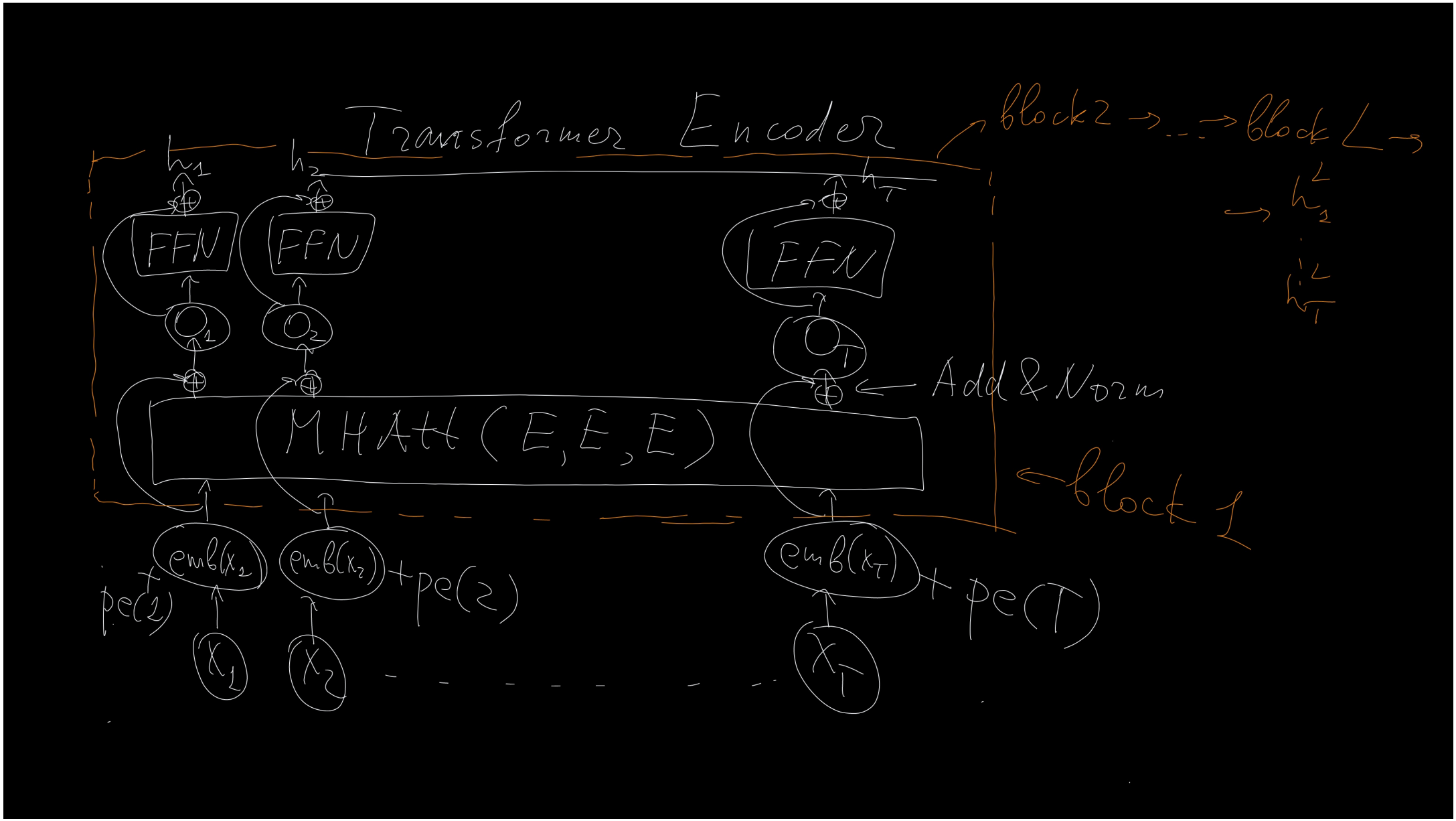
Multi Head Attention

$$head_i = Att(\underbrace{Q}_{\tilde{T}\times H} \underbrace{W_i^Q}_{H\times H_{head}}, \underbrace{K}_{H\times H_{head}} \underbrace{W_i^K}, \underbrace{V}_{H\times H_{head}} \underbrace{W_i^V}_{H\times H_{head}})$$

$$\underbrace{O}_{\tilde{T}\times H} = MHAtt = concat\underbrace{[head_1, \dots, head_h]}_{\tilde{T}\times H_{head}} \underbrace{W^O}_{hH_{head}\times H}$$

$\{W_i^K, W_i^Q, W_i^V\}_{i=1}^{h}, W^O$ – learnable parameters

$\tilde{T}\times hH_{head}$


Transformer Encoder
block 2 → ... → block L →
h_1
h_2
h_T
FFN
FFN
FFN
O_1
O_2
O_T
Add & Norm
MHAtt(E, E, E)
block 1
emb(x_1) + pe(1)
emb(x_2) + pe(2)
emb(x_T) + pe(T)
x_1
x_2
x_T

$h_t^0 = emb(x_t), \quad t = 1, \ldots, T$

for layer $l = 1, \ldots, L$:

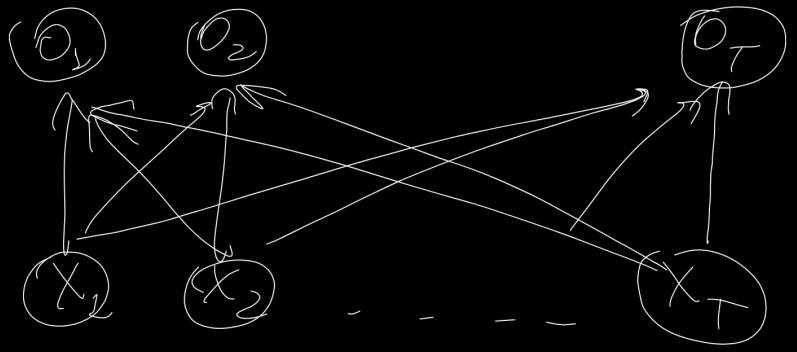
$$O^l = LayerNorm\left(H^{l-1} + MHAtt\left(H^{l-1}, H^{l-1}, H^{l-1}\right)\right)$$

$$H^l = LayerNorm\left(O^l + g\left(O^l W_1 + b_1\right) W_2 + b_2\right)$$

$T \times H \quad H \times 4H \quad 4H \times H$

act. function

Output: H^L



$$h_{t+1} = g(\underset{H\times H}{W_x} \underset{H\times 1}{x_t} + W_h h_t + b)$$

quadratic depend. on T

T, H, H_{head}, h

RNN	MHAtt
$\underline{O}(T \cdot H \cdot H)$	$\underline{O}(h(T \cdot H \cdot H_{head} + T \cdot T \cdot H_{head})) = \{hH_{head} = H\} =$
linear dep. on T	$= T \cdot H \cdot H + T \cdot T \cdot H$

Positional Embedding

$pe_t \in \mathbb{R}^d$

$$pe_{t,2i} = \sin(\theta_i t)$$
$$pe_{t,2i+1} = \cos(\theta_i t)$$
$$i = 0, \ldots, \frac{d}{2}$$
$$\theta_i = 10000^{-\frac{2i}{d}}$$

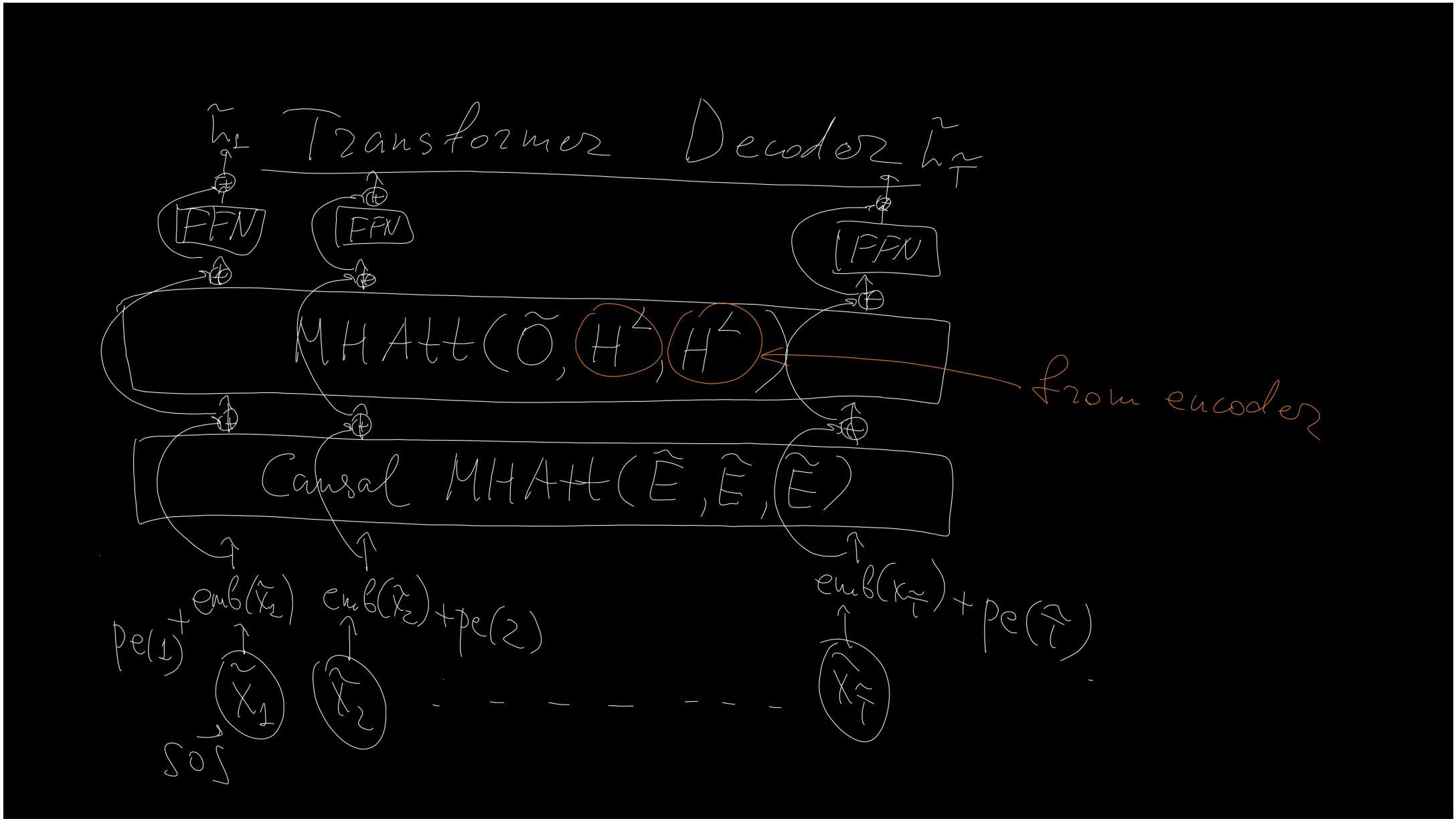
$$pe_{t+k,2i} = \sin(\theta_i(t+k)) =$$

$i=0 \Rightarrow \theta_i = 1$

$i = \frac{d}{2} \Rightarrow \theta_i = \frac{1}{10000}$

$$= \underbrace{\sin(\theta_i t)}_{pe_{t,2i}} \cos(\theta_i k) + \underbrace{\cos(\theta_i t)}_{pe_{t,2i+1}} \sin(\theta_i k) =$$

$$= \text{Linear}(pe_{t,2i}, pe_{t,2i+1})$$


Transformer Decoder
FFN
FFN
FFN
MHAtt(Õ, H, H)
from encoder
Causal MHAtt(Ẽ, Ẽ, Ẽ)
emb(x̃1) + pe(1)
emb(x̃2) + pe(2)
emb(x̃T̃) + pe(T̃)
x̃1
x̃2
x̃T̃
SOS

$$\tilde{h}_t^0 = emb(x_t), \quad t = 1, \ldots, T$$

for $l = 1, 2, \ldots, \tilde{L}$:

$$O^l = LayerNorm\left(\tilde{H}^{l-1} + CausalMHAtt(\tilde{H}^{l-1}, \tilde{H}^{l-1}, \tilde{H}^{l-1})\right)$$

$$O^l = LayerNorm\left(O^l + MHAtt(O^l, H^L, H^L)\right)$$

$$\tilde{H}^l = LayerNorm\left(O^l + g(O^l W_1 + b_1) W_2 + b_2\right)$$

Output: $\tilde{H}^{\tilde{L}}$

$$O = \text{CausalAtt}(Q, K, V) =$$

$$= \text{Softmax}\left(\frac{QK^T}{\sqrt{H}} + M\right)V$$

$$M = \begin{bmatrix} 0 & -\infty & -\infty & \cdots & -\infty \\ 0 & 0 & -\infty & \cdots & -\infty \\ 0 & 0 & 0 & \ddots & \vdots \\ \vdots & \vdots & \ddots & \ddots & -\infty \\ 0 & 0 & \cdots & 0 & 0 \end{bmatrix} \in \mathbb{R}^{T \times T}$$

KV cache

Rotational Pos. Emb (RoPE)

- $\langle R_m q, R_n k \rangle$ – should depend on R_{n-m}

$q, k \in \mathbb{R}^2$ $\quad R_{\theta,m} = \begin{bmatrix} \cos(\theta_m) & -\sin(\theta_m) \\ \sin(\theta_m) & \cos(\theta_m) \end{bmatrix}$

$R_{\theta,m} q$ – rotation of q by angle θ_m

$\langle R_{\theta,m} q, R_{\theta,n} k \rangle = q^T R_{\theta,m}^T R_{\theta,n} k = q^T R_{\theta,n-m} k$

$$\begin{bmatrix} \tilde{k}_{2i} \\ \tilde{k}_{2i+1} \end{bmatrix} = R_{\theta_i, m} \begin{bmatrix} k_{2i} \\ k_{2i+1} \end{bmatrix}, \quad i = 0, \dots, \frac{d}{2}$$

$$\theta_i = 10000^{-\frac{2i}{d}}$$

$$i = 0 \quad \theta_i = 1$$

$$i = \frac{d}{2} \quad \theta_i = \frac{1}{10000}$$

Dynamic NTK scaling

$$\theta_i = 10000^{-\frac{2i}{d}}$$

$$\theta_i = \left(10000 \cdot s^{\frac{d}{d-2}}\right)^{-\frac{2i}{d}}, \quad s = \max\left(1, \frac{\text{input text length}}{\text{train text length}}\right)$$

$i = 0, \quad \theta_i = 1$

$i = \frac{d}{2}, \quad \theta_i = \frac{1}{10000 \cdot s^{\frac{d}{d-2}}}$

Gated FFN

$$\underset{T\times H}{O} = \Big(\mathrm{SiLu}\big(\underset{T\times H}{X}\ \underset{H\times H_1}{W_{gate}}\big) \odot \big(\underset{H\times H_2}{X W_{up}}\big)\Big)\ \underset{H_1\times H}{W_{down}}$$

$$H_1 = \frac{8}{3} H$$

Modern Transformer architecture

- RoPE on every layer, Dynamic NTK scaling, no pe on inputs
- Gated FFN instead of FFN, no biases
- RMSnorm instead of LayerNorm
- KV cache for inference in autoregr. decoder
- FlashAttention for GPU-aware fast att. computation